DEPARTMENT OF SOFTWARE ENGINEERING
Faculty of Computing & Information Technology (FCIT)
University of the Punjab, Lahore – PAKISTAN



FYDP-DSE Proposal template

[BSEF23-02]

[RobotEshop]

Bachelors of Science in Software Engineering(2023-2027)

By

Abdul Mateen Butt	BSEF23M036
Hamna Hameed	BSEF23M007
Alishba Abid	BSEF23M021
Muhammad Umer Farooq	BSEF23M004

Supervised by:

Dr. Fatima Sabir
«Primary_Designation»

«CoSupervisor»,
«CoDesignation»

**FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY,
UNIVERSITY OF THE PUNJAB, LAHORE.**

Proposal Approval Signatures (FAC)

We, the undersigned members of the assigned Faculty Advisory Committee (FAC), have reviewed the submitted project proposal. Based on our initial assessment, we acknowledge the group's submission and confirm that the proposal has been evaluated for completeness and alignment with FYDP objectives.

Group ID: BSEF23-02

Project title: RobotEshop

Primarysupervisor: Dr. Fatima Sabir

(Signature)

Co-supervisor: «CoSupervisor»

(Signature)

Signatures(Faculty Advisory Committee(FAC)

	FAC member#1	FAC member#2
Name	«Evaluator1_name»	«Evaluator2_name»
Signature		

Head of FYDP Coordination Office: Dr Natalia Chaudhry

(Signature)



Content

S

1. Project Title and abstract.....
2. Team Details.....
3. Problem Statement / Motivation.....
4. Project Objectives.....
5. System Overview.....
6. Development Methodology.....
7. Technology Stack.....
8. Project Timeline / Milestones.....
Milestone.....
Expected Deadline.....
9. Market Fit / Commercial Potential.....
10. Competitor Analysis.....

1. Project Title and abstract

Project Title: RobotEshop - A Unified Web Aggregation Platform for Educational and Industrial Robots

Abstract: Finding the right programmable robot is difficult because information is scattered across multiple websites, with missing specifications and technical jargon that beginners struggle to understand. General-purpose chatbots often provide inaccurate recommendations, suggest robots that are unavailable in Pakistan, or fail to link users to actual products and learning resources. This project solves these issues by offering a dedicated platform with verified robot information, local availability, programming guides, tutorial videos, and direct purchasing options.

2. Team Details

Name	Roll Number	Email	Role
Abdul Mateen Butt	BSEF23M036	bsef23m036@pucit.edu.pk	Developer
Hamna Hameed	BSEF23M007	bsef23m007@pucit.edu.pk	Scrum Master
Alishba Abid	BSEF23M021	bsef23m021@pucit.edu.pk	Developer
Muhammad Umer Farooq	BSEF23M004	bsef23m004@pucit.edu.pk	Developer

3. Problem Statement / Motivation

The global robotics market is experiencing exponential growth, yet the purchasing process for these high-value assets remains highly fragmented. Prospective buyers—ranging from academic researchers to industrial procurement officers—are currently forced to navigate a disjointed ecosystem of manufacturer websites, third-party e-commerce giants (like Amazon and Alibaba), and specialized B2B industrial directories. This fragmentation creates significant inefficiencies: technical specifications are often presented in inconsistent formats, real-time price comparisons are nearly impossible, and the verification of authorized distributors requires manual cross-referencing. These obstacles lead to prolonged procurement cycles, missed opportunities for cost savings, and an increased risk of purchasing from unverified suppliers. Therefore, there is a pressing need for a centralized, intelligent aggregation system that consolidates this scattered data into a single, trustworthy point of access.

4. Project Objectives

- To develop a centralized robotics marketplace that gathers robot information from different sources in one place.
- To provide complete details for each robot, including hardware and software specifications, supported platforms, and programming tools.
- To allow users to search, filter, and compare robots easily based on their needs.
- To show trusted and verified sellers so users can buy with confidence.
- To reduce the time, effort, and confusion involved in finding the right robot.



- To support age-based and skill-based recommendations for educational robots by clearly identifying suitable user groups.
- To provide the feature of price comparison between different robots.
- To provide the information about the operating system supported.

5. System Overview

The proposed system is a centralized robotics information platform designed to collect, organize, and present robot-related data from multiple vendor and marketplace sources. The backend will use automated web scraping agents, implemented with Python-based tools such as Scrapy, to periodically crawl selected websites and extract key attributes including hardware specifications, software requirements, platform compatibility, programming support, and product pricing. The extracted data will then be cleaned, normalized, and stored in a relational database such as PostgreSQL to ensure consistency and efficient querying.

On the frontend, a modern JavaScript framework such as React will be used to build an interactive and user-friendly interface that supports search, filtering, side-by-side comparison, and historical price tracking. The system will also provide compatibility details for platforms such as Windows, Linux, Arduino, Raspberry Pi, ROS, and mobile ecosystems, along with recommended IDEs or software tools required to program or control each robot. These modules will work together to give users a single standardized view of the robotics market and help them make faster, more informed purchasing decisions. Docker will also be added for higher portability.

6. Development Methodology

The proposed project will follow the **Agile (Scrum)** methodology. This approach is suitable because the system consists of multiple interconnected features, such as data scraping, normalization, database management, search, filtering, and comparison, which can be developed in incremental stages. Agile allows the project team to adapt to changing requirements, receive supervisor feedback regularly, and continuously improve the system throughout development.

7. Technology Stack

List all the technologies, frameworks, and tools that will be used in different layers of your project:

Layer	**Technologies/Tools**
Frontend	React, HTML5, CSS3, Javascript



Backend	Django, Fast API's, TypeScript
Database	PostgreSQL
Dev Tools	Git, GitHub

8. Project Timeline / Milestones

Milestone	Expected Deadline
Requirements Gathering & Finalization	Month 1
System Design & Prototype Development	Month 2
Frontend, Backend & AI Development	Month 3 to 6
Testing & Bug Fixing	Month 7
Final Deployment, Documentation & Presentation	Month 8

9. Market Fit / Commercial Potential

RobotEshop addresses a real need in the robotics market by bringing scattered product information into one centralized platform. At present, users such as students, researchers, and procurement teams must visit multiple websites to compare robots, check specifications, and verify sellers. This process is time-consuming, confusing, and often unreliable. By offering a single place to search, compare, and evaluate robotics products, RobotEshop can save users time and help them make better purchasing decisions.

The platform is unique because it is not just a regular e-commerce website. It focuses specifically on robotics products and provides useful features such as technical specifications, compatibility details, programming tools, age-based recommendations, and trusted seller information. These features make it more valuable than general marketplaces, especially for educational and professional users who need accurate technical information before buying.

From a commercial point of view, RobotEshop has strong potential because it can attract a wide range of users, including individuals, schools, universities, robotics labs, and small businesses. The platform can be monetized through affiliate links, sponsored listings, featured product placements, and partnerships with verified vendors. In the future, it can also expand into premium services such as advanced comparison tools, price alerts, and vendor analytics.

10. Competitor Analysis

In Pakistan, robotics e-commerce websites such as **Robo.pk, RoboticPK, and IndusRoboTix** provide online platforms for browsing and purchasing programmable robots. However, most of these websites do not offer a dedicated AI chatbot to guide customers in selecting the most suitable robot or to explain technical specifications in simple terms. On the other hand, general e-commerce platforms such as **Daraz** include customer support or AI-powered chat features to assist users with navigation and order-related queries, but these chatbots are not specialized in robotics and cannot provide detailed product guidance or personalized robot recommendations. This highlights a significant gap in the market, which our proposed application aims to address by offering a dedicated AI assistant focused exclusively on programmable robots, their specifications, comparisons, and purchase guidance.

This paragraph clearly distinguishes:

Websites without a chatbot: Robo.pk, RoboticPK, IndusRoboTix.

Website with a chatbot/customer assistant: Daraz (for shopping assistance, not robot guidance).

Website

Gap: No dedicated AI chatbot exists for helping users understand and choose programmable robots.

Robotics e-commerce websites are not providing video tutorials for users to tell them “How to use it?”. And it produces multiple problems at the time of usage like waste of time and effort. So to fill this gap we will provide the tutorials of the robots on RoboEShop. And it will help users a lot.